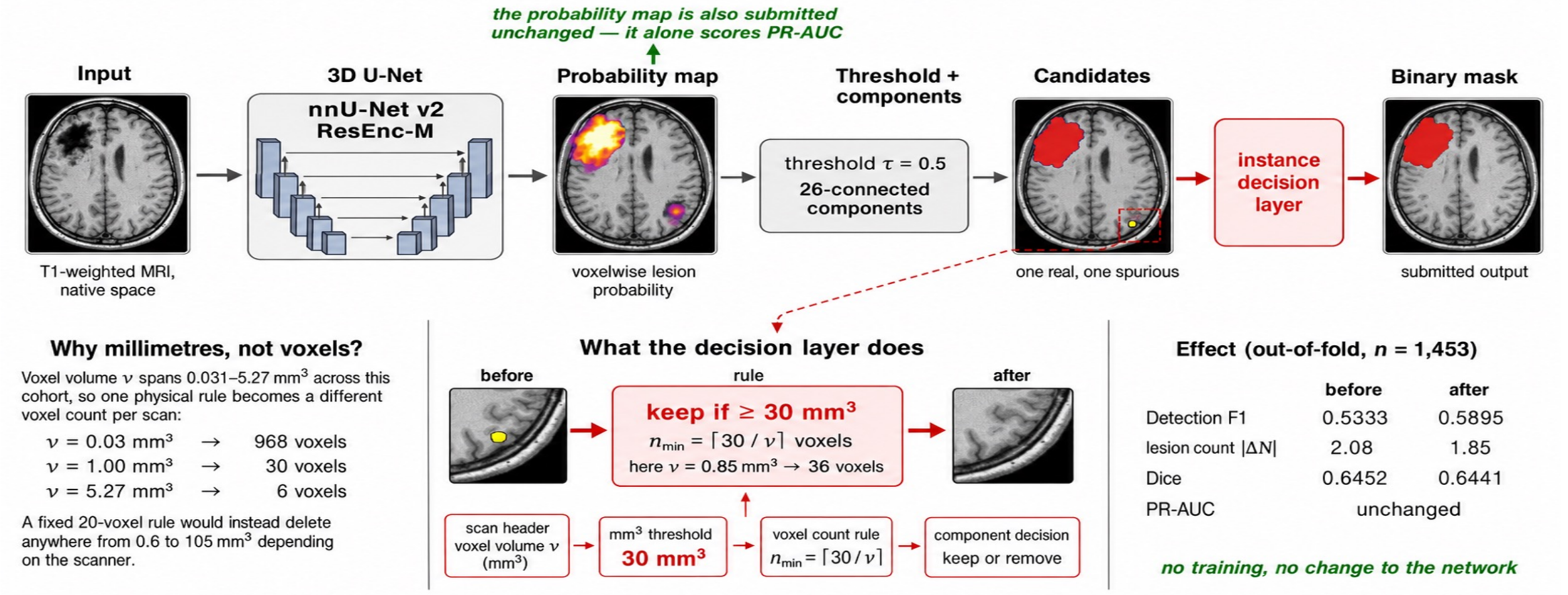

the probability map is also submitted unchanged — it alone scores PR-AUC
Input
3D U-Net
nnU-Net v2 ResEnc-M
Probability map
Threshold + components
threshold τ = 0.5
26-connected components
Candidates
instance decision layer
Binary mask
T1-weighted MRI, native space
voxelwise lesion probability
one real, one spurious
submitted output
Why millimetres, not voxels?
Voxel volume ν spans 0.031–5.27 mm³ across this cohort, so one physical rule becomes a different voxel count per scan:
ν = 0.03 mm³ → 968 voxels
ν = 1.00 mm³ → 30 voxels
ν = 5.27 mm³ → 6 voxels
A fixed 20-voxel rule would instead delete anywhere from 0.6 to 105 mm³ depending on the scanner.
What the decision layer does
before
rule
after
keep if ≥ 30 mm³
n_min = ⌈30 / ν⌉ voxels
here ν = 0.85 mm³ → 36 voxels
scan header voxel volume ν (mm³)
mm³ threshold 30 mm³
voxel count rule n_min = ⌈30 / ν⌉
component decision keep or remove
Effect (out-of-fold, n = 1,453)
before after
Detection F1 0.5333 0.5895
lesion count |ΔN| 2.08 1.85
Dice 0.6452 0.6441
PR-AUC unchanged
no training, no change to the network